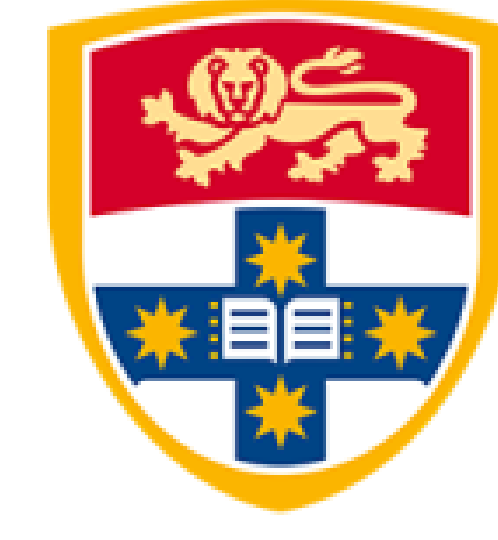


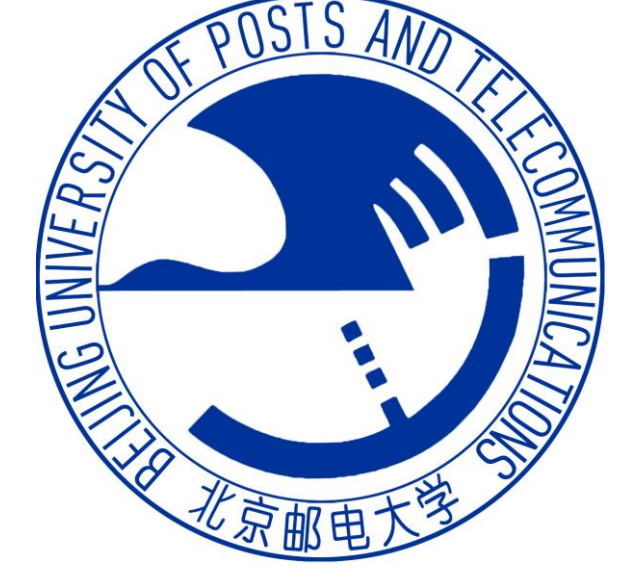
Mitigating Legal Hallucinations via Symbolic Constraints and Analogical Precedents

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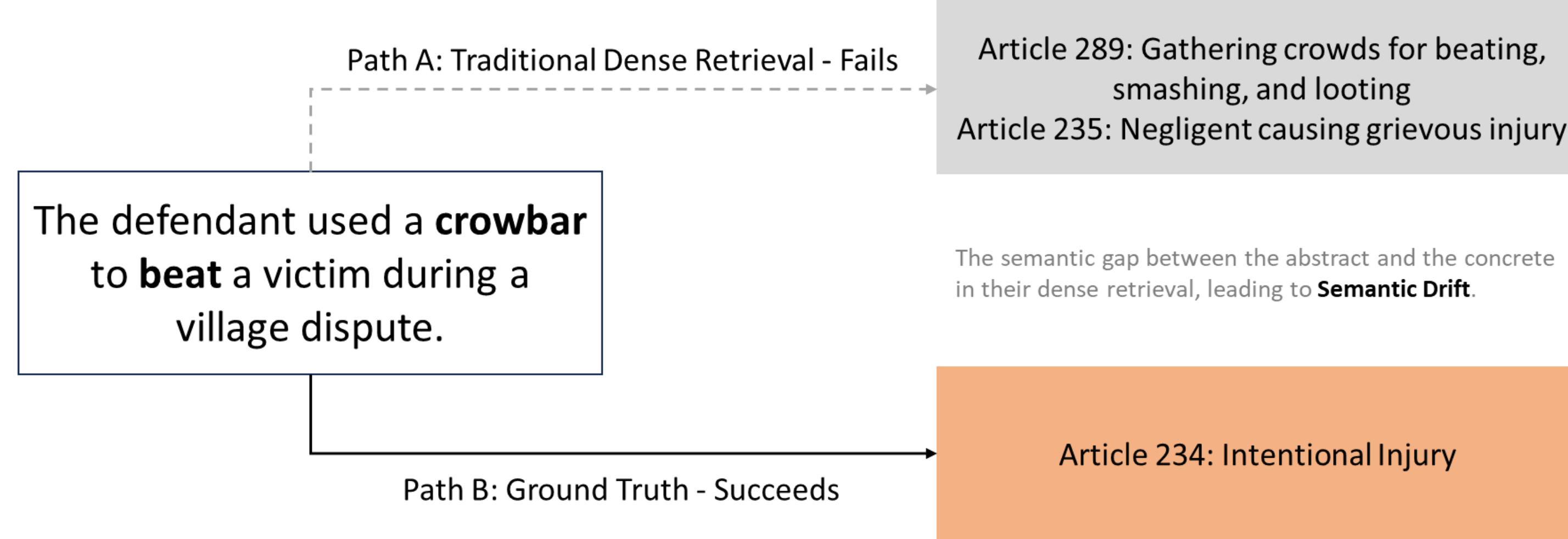
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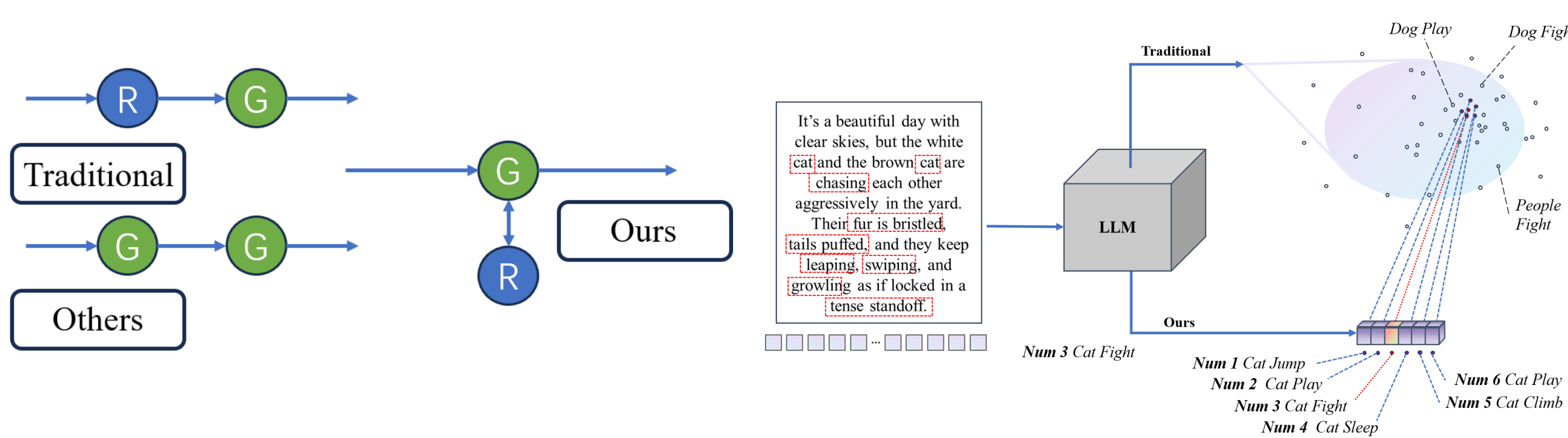
❖ Background & Problem



- **Hallucinations in Legal LLMs:** Current legal LLMs inherently struggle with complex terminology and rigorous logical structures, leading to substantial hallucination risks during legal analysis.
- **Semantic Drift:** Standard dense retrieval strategies fail to bridge the semantic gap between concrete factual descriptions and abstract legal norms, resulting in semantic drift.
- **Static Retrieval Limitations:** Traditional fixed top-k and threshold-based retrieval strategies fail to adapt to varying case complexities, often introducing irrelevant noise or missing ground truths.

❖ Core Method

➤ I. Symbolic Constrained Retrieval (SCR)

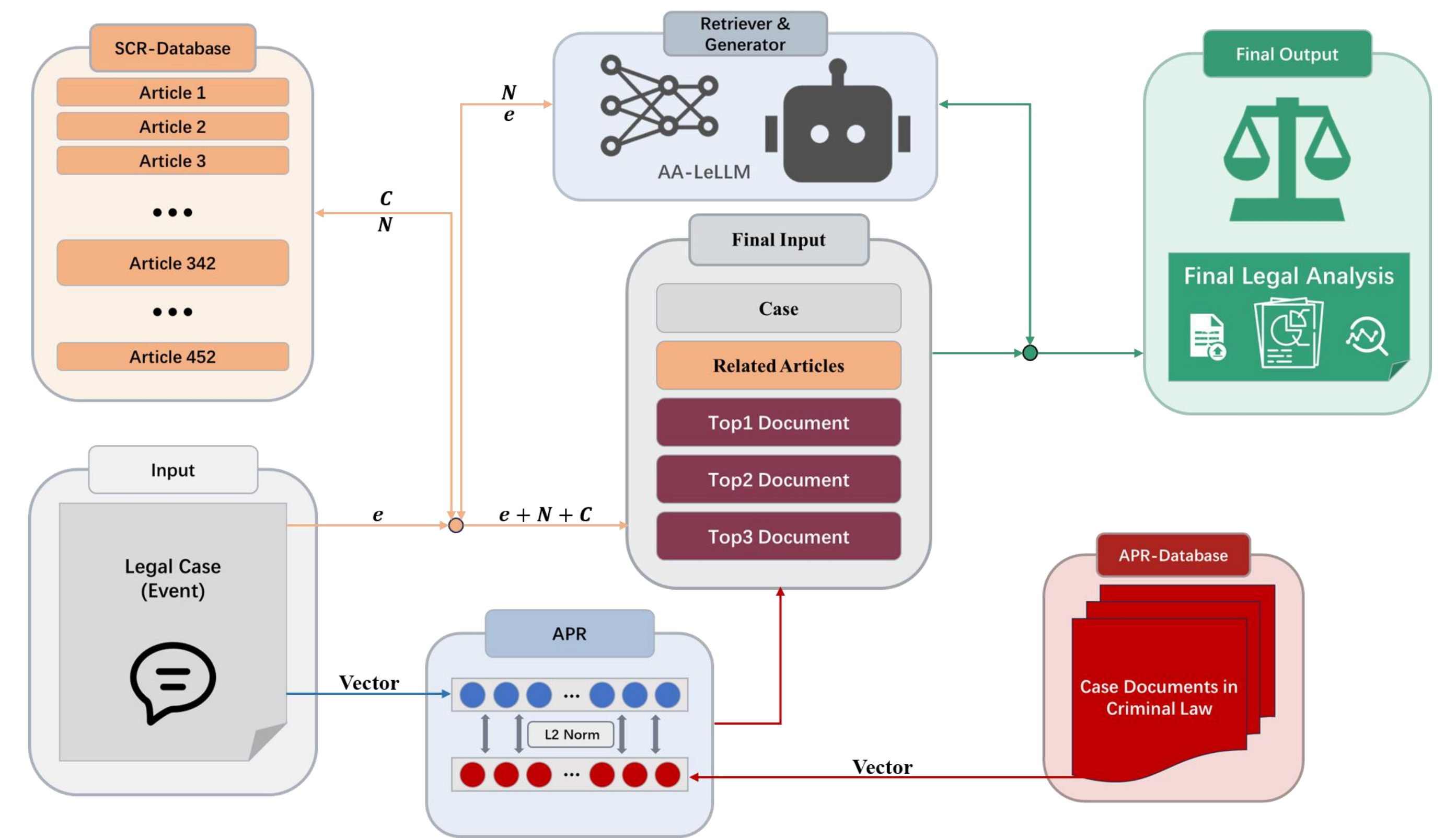


- **Lemma 1: Hallucination Risk Upper Bound**
$$1 - \text{Acc}(\theta_0) \leq 1 - \text{Acc}(\theta) \cdot \text{Auth}(\theta)$$
- **Mechanism:** Formulates legal citation as a closed-set symbolic prediction task by parametrizing retrieval directly into the LLM backbone.
- **Hallucination Mitigation:** Minimizes hallucination risks by grounding generative outputs in verifiable, low-dimensional symbolic identifiers.
- **Adaptive Citation:** Enables the model to dynamically determine the precise number of cited articles based on inherent case complexity via a learned dynamic threshold, bypassing rigid top-k rules: $\hat{N} = \{n \mid p_{\theta}^{\text{cls}}(x)[n] \geq \tau_{\theta}^*(x)\}$

➤ II. Analogical Precedent Retrieval (APR)

- **Mechanism:** Utilizes dense retrieval over a newly collected, large-scale dataset of 176k criminal court cases to fetch semantically analogous judgments.
- Mitigates reasoning drift by aligning factual analysis with verifiable open-set precedents, substantially enhancing the explainability of the final decision.

❖ Full Architecture



➤ Theoretical Alignment: Legal Syllogism

- **Major Premise (Legal Norms):** SCR strictly retrieves constrained closed-set legal articles.
- **Minor Premise (Fact Analysis):** APR retrieves open-set analogous precedents for reliable reference.
- **Conclusion (Legal Analysis):** AA-LeLLM synthesizes both to generate explainable judgments.

❖ Empirical Validation

➤ SOTA Performance & Reduced Hallucinations

- Our method beats baselines.
- New **HR-Benchmark**.

Model	Criminal		Full Domain				Average
	FAP	CP	DFI	MDI	ITI	ED	
DeepSeek-7B (Guo et al., 2025)	16.86	28.89	27.20	39.69	35.80	58.35	34.47
LawGPT-beta1.1-7B (Zhou et al., 2024)	0.15	15.68	4.95	6.85	2.40	14.94	7.50
LexiLaw-6B (Li et al., 2024)	13.15	39.99	3.30	15.60	22.80	15.30	18.36
HanFei-7B (He et al., 2023)	2.64	30.96	6.39	30.44	30.20	14.73	19.23
Wisdom-Interrogatory-7B (Wu et al., 2024)	32.84	35.09	7.84	36.72	21.00	15.98	24.91
Fuzi-Mingcha-7B (Deng et al., 2023)	25.19	55.93	19.59	28.46	18.60	16.90	27.45
Qwen3-8B (Team, 2025)	73.21	51.88	46.00	54.40	38.80	65.86	55.02
Internlm3-8B (Cai et al., 2024)	82.92	55.02	37.20	52.29	41.20	66.38	55.83
AA-LeLLM-7B(Ours)	88.84 ± 0.34	57.93 ± 0.60	20.10 ± 0.50	46.29 ± 1.19	39.00 ± 1.52	81.36 ± 0.07	55.59

Category	Method	Parameter	F1-Score	Precision	Recall
Dense Retrieval	BGE-v1.5 (Xiao et al., 2024)	$k^{\dagger} = 1$	24.26	26.60	18.19
		$\tau^{\dagger} = 0.97$	16.73	10.92	31.33
	BGE-m3 (Chen et al., 2024)	$k^{\dagger} = 1$	52.68	59.40	40.63
		$\tau^{\dagger} = 0.56$	36.77	25.89	55.95
Sparse Retrieval	SAILER (Li et al., 2023)	$k^{\dagger} = 1$	63.63	72.20	49.38
		$\tau^{\dagger} = 0.63$	48.49	34.10	59.10
Generative Retrieval	BM25	$k^{\dagger} = 1$	35.19	40.20	27.50
		$k = 2$	30.36	25.50	34.88
		$k = 3$	26.37	19.33	39.67
Generative Retrieval	SCR (Qwen3-8B)	$\tau_{\theta}^*(x)$	73.20	67.70	74.83
		SCR (Internlm3-8B)	82.92	43.81	78.39
		SCR (AALawyer)	88.94	89.67	84.26

Baseline	Method	Hallu↓	Prof	Info	Expa	Avg.	Baseline	Method	Avg.Acc	Avg.Auth	H(θ)↓
DS-7B	Vanilla	82.4	58.8	40.0	44.6	40.2	DS-7B	Vanilla	0.29	0.44	0.88
	SCR	85.4	46.4	39.4	44.2	36.2		SCR	0.59	0.70	0.58
	APR	63.0	66.6	81.6	78.8	66.0	Qwen3-8B	Vanilla	0.59	0.94	0.44
	AALawyer	81.4	55.0	74.0	70.8	54.6		SCR	0.71	0.69	0.51
AA-LeLLM	Vanilla	57.0	61.9	24.8	40.7	42.6	Internlm3-8B	Vanilla	0.71	0.95	0.33
	SCR	39.6	70.6	37.9	67.8	59.1		SCR	0.76	0.49	0.63
	APR	53.9	60.9	74.7	79.5	65.3	AA-LeLLM	Vanilla	0.76	0.93	0.29
	AALawyer	44.8 ± 1.8	72.2 ± 0.2	81.9 ± 0.2	88.0 ± 0.2	74.3		SCR			